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Semiconductor device and method for fabricating the same

Abstract

A semiconductor device includes a magnetic tunneling junction (MTJ) on a substrate, a first spacer on one side of the of the MTJ, a second spacer on another side of the MTJ, a first metal interconnection on the MTJ, and a liner adjacent to the first spacer, the second spacer, and the first metal interconnection. Preferably, each of a top surface of the MTJ and a bottom surface of the first metal interconnection includes a planar surface and two sidewalls of the first metal interconnection are aligned with two sidewalls of the MTJ.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/341,417, filed on Jun. 8, 2021, which is a continuation application of U.S. application Ser. No. 16/438,480, filed on Jun. 12, 2019. The contents of these applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to a semiconductor device and method for fabricating the same, and more particularly to a magnetoresistive random access memory (MRAM) and method for fabricating the same.

2. Description of the Prior Art

(2) Magnetoresistance (MR) effect has been known as a kind of effect caused by altering the resistance of a material through variation of outside magnetic field. The physical definition of such effect is defined as a variation in resistance obtained by dividing a difference in resistance under no magnetic interference by the original resistance. Currently, MR effect has been successfully utilized in production of hard disks thereby having important commercial values. Moreover, the characterization of utilizing GMR materials to generate different resistance under different magnetized states could also be used to fabricate MRAM devices, which typically has the advantage of keeping stored data even when the device is not connected to an electrical source.

(3) The aforementioned MR effect has also been used in magnetic field sensor areas including but not limited to for example electronic compass components used in global positioning system (GPS) of cellular phones for providing information regarding moving location to users. Currently, various magnetic field sensor technologies such as anisotropic magnetoresi stance (AMR) sensors, GMR sensors, magnetic tunneling junction (MTJ) sensors have been widely developed in the market. Nevertheless, most of these products still pose numerous shortcomings such as high chip area, high cost, high power consumption, limited sensibility, and easily affected by temperature variation and how to come up with an improved device to resolve these issues has become an important task in this field.

SUMMARY OF THE INVENTION

(4) According to an embodiment of the present invention, a semiconductor device includes a magnetic tunneling junction (MTJ) on a substrate, a first spacer on one side of the of the MTJ, a second spacer on another side of the MTJ, a first metal interconnection on the MTJ, and a liner adjacent to the first spacer, the second spacer, and the first metal interconnection. Preferably, each of a top surface of the MTJ and a bottom surface of the first metal interconnection includes a planar surface and two sidewalls of the first metal interconnection are aligned with two sidewalls of the MTJ.

(5) According to another aspect of the present invention, a semiconductor device includes a magnetic tunneling junction (MTJ) on a substrate, a first liner adjacent to the MTJ, a second liner on the first liner, and a first metal interconnection on the MTJ. Preferably, each of the top surface of the MTJ and a bottom surface of the first metal interconnection includes a planar surface, two sidewalls of the first metal interconnection are aligned with two sidewalls of the MTJ, and the first liner and the second liner are made of different materials.

(6) According to yet another aspect of the present invention, a semiconductor device includes a magnetic tunneling junction (MTJ) on a substrate, a first liner adjacent to the MTJ, a second liner on the first liner, and a first metal interconnection on the MTJ. Preferably, the first metal interconnection includes protrusions adjacent to two sides of the MTJ, a bottom surface of the protrusions contact the first liner and the second liner directly, and a sidewall of the first metal interconnection is aligned with a sidewall of the second liner.

(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1-6** illustrate a method for fabricating a semiconductor device according to an embodiment of the present invention.

(2) FIG. **7** illustrates a structural view of a semiconductor device according to an embodiment of the present invention.

(3) FIG. **8** illustrates a structural view of a semiconductor device according to an embodiment of the present invention.

(4) FIG. **9** illustrates a structural view of a semiconductor device according to an embodiment of the present invention.

(5) FIG. **10** illustrates a structural view of a semiconductor device according to an embodiment of the present invention.

(6) FIG. **11** illustrates a structural view of a semiconductor device according to an embodiment of the present invention.

DETAILED DESCRIPTION

(7) Referring to FIGS. **1-6**, FIGS. **1-6** illustrate a method for fabricating a semiconductor device, or more specifically a MRAM device according to an embodiment of the present invention. As shown in FIG. **1**, a substrate **12** made of semiconductor material is first provided, in which the semiconductor material could be selected from the group consi sting of silicon (Si), germanium (Ge), Si—Ge compounds, silicon carbide (SiC), and gallium arsenide (GaAs), and a MTJ region **14** and a logic region **16** are defined on the substrate **12**.

(8) Active devices such as metal-oxide semiconductor (MOS) transistors, passive devices, conductive layers, and interlayer dielectric (ILD) layer **18** could also be formed on top of the substrate **12**. More specifically, planar MOS transistors or non-planar (such as FinFETs) MOS transistors could be formed on the substrate **12**, in which the MOS transistors could include transistor elements such as gate structures (for example metal gates) and source/drain region, spacer, epitaxial layer, and contact etch stop layer (CESL). The ILD layer **18** could be formed on the substrate **12** to cover the MOS transistors, and a plurality of contact plugs could be formed in the ILD layer **18** to electrically connect to the gate structure and/or source/drain region of MOS transistors. Since the fabrication of planar or non-planar transistors and ILD layer is well known to those skilled in the art, the details of which are not explained herein for the sake of brevity.

(9) Next, metal interconnect structures **20**, **22** are sequentially formed on the ILD layer **18** on the MTJ region **14** and the edge region **16** to electrically connect the aforementioned contact plugs, in which the metal interconnect structure **20** includes an inter-metal dielectric (IMD) layer **24** and metal interconnections **26** embedded in the IMD layer **24**, and the metal interconnect structure **22** includes a stop layer **28**, an IMD layer **30**, and metal interconnections **32** embedded in the stop layer **28** and the IMD layer **30**.

(10) In this embodiment, each of the metal interconnections **26** from the metal interconnect structure **20** preferably includes a trench conductor and each of the metal interconnections **32** from the metal interconnect structure **22** on the MTJ region **14** includes a via conductor. Preferably, each of the metal interconnections **26**, **32** from the metal interconnect structures **20**, **22** could be embedded within the IMB layers **24**, **30** and/or stop layer **28** according to a single damascene process or dual damascene process. For instance, each of the metal interconnections **26**, **32** could further includes a barrier layer **34** and a metal layer **36**, in which the barrier layer **34** could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer **36** could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). Since single damascene process and dual damascene process are well known to those skilled in the art, the details of which are not explained herein for the sake of brevity. In this embodiment, the metal layers **36** are preferably made of copper, the IMD layers **24**, **30** are

preferably made of silicon oxide, and the stop layers **28** is preferably made of nitrogen doped carbide (NDC), silicon nitride, silicon carbon nitride (SiCN), or combination thereof.

(11) Next, a MTJ stack **38** or stack structure is formed on the metal interconnect structure **22**, a cap layer **40** is formed on the MTJ stack **38**, and another cap layer **42** formed on the cap layer **40**. In this embodiment, the formation of the MTJ stack **38** could be accomplished by sequentially depositing a first electrode layer **44**, a fixed layer **46**, a free layer **48**, a capping layer **50**, and a second electrode layer **52** on the IMB layer **30**. In this embodiment, the first electrode layer **44** and the second electrode layer **52** are preferably made of conductive material including but not limited to for example Ta, Pt, Cu, Au, Al, or combination thereof. The fixed layer **46** could be made of antiferromagnetic (AFM) material including but not limited to for example ferromanganese (FeMn), platinum manganese (PtMn), iridium manganese (IrMn), nickel oxide (NiO), or combination thereof, in which the fixed layer **46** is formed to fix or limit the direction of magnetic moment of adjacent layers. The free layer **48** could be made of ferromagnetic material including but not limited to for example iron, cobalt, nickel, or alloys thereof such as cobalt-iron-boron (CoFeB), in which the magnetized direction of the free layer **48** could be altered freely depending on the influence of outside magnetic field. The capping layer **50** could be made of insulating material including but not limited to for example oxides such as aluminum oxide (AlO.sub.x) or magnesium oxide (MgO). Preferably, the cap layer **40** and cap layer **42** are made of different materials. For instance, the cap layer **40** is preferably made of silicon nitride and the cap layer **42** is made of silicon oxide, but not limited thereto.

(12) Next, a patterned mask **54** is formed on the cap layer **42**. In this embodiment, the patterned mask **54** could include an organic dielectric layer (ODL) **56**, a silicon-containing hard mask bottom anti-reflective coating (SHB) **58**, and a patterned resist **60**.

(13) Next, as shown in FIG. **2**, one or more etching process is conducted by using the patterned mask **54** as mask to remove part of the cap layers **40**, **42**, part of the MTJ stack **38**, and part of the IMD layer **30** to form a MTJ **62** on the MTJ region **14**, in which the first electrode layer **44** at this stage preferably becomes a bottom electrode **76** for the MTJ **62** while the second electrode layer **52** becomes a top electrode **78** for the MTJ **62** and the cap layers **40**, **42** could be removed during the etching process. It should be noted that this embodiment preferably conducts a reactive ion etching (RIE) process by using the patterned mask **54** as mask to remove part of the cap layers **40**, **42** and part of the MTJ stack **38**, strips the patterned mask **54**, and then conducts an ion beam etching (IBE) process by using the patterned cap layer **42** as mask to remove part of the MTJ stack **38** and part of the IMD layer **30** to form MTJ **62**. Due to the characteristics of the IBE process, the top surface of the remaining IMD layer **30** is slightly lower than the top surface of the metal interconnections **32** after the IBE process and the top surface of the IMD layer **30** also reveals a curve or an arc.

(14) It should also be noted that when the IBE process is conducted to remove part of the IMD layer **30**, part of the metal interconnection **32** is removed at the same time so that a first slanted sidewall **64** and a second slanted sidewall **66** are formed on the metal interconnection **32** adjacent to the MTJ **62**, in which each of the first slanted sidewall **64** and the second slanted sidewall **66** could further include a curve (or curved surface) or a planar surface. Moreover, if the second electrode layer **52** were made of tantalum (Ta), more second electrode layer **52** closer to the tip of the MTJ **62** is preferably removed during the patterning of the MTJ stack **38** through the IBE process so that inclined sidewalls and top surfaces are formed on the patterned MTJ **62**. Specifically, the tip or top portion the top electrode **78** of the MTJ **62** formed at the stage preferably includes a reverse V-shape or a curve (not shown) while two sidewalls of the MTJ **62** are slanted sidewalls.

(15) Next, as shown in FIG. **3**, a liner **68** is formed on the MTJ **62** to cover the surface of the IMB layer **30**. In this embodiment, the liner **68** is preferably made of silicon nitride (SiN), but could also be made of other dielectric material including but not limited to for example silicon oxide, silicon

oxynitride, or silicon carbon nitride.

(16) Next, as shown in FIG. **4**, an etching process is conducted to remove part of the liner **68** to form a spacer including a first spacer **70** and a second spacer **82** adjacent to the MTJ **62**, in which the first spacer **70** is disposed on a sidewall of the MTJ **62** to cover and contact the first slanted sidewall **64** of the metal interconnection **32** and the second spacer **82** and the second spacer **82** is disposed on another sidewall of the MTJ **62** to cover an contact the second slanted sidewall **66**.

(17) Next, an atomic layer deposition (ALD) process is conducted to form a second liner **100** to cover the surfaces of the IMD layer **30**, the MTJ **62**, the first spacer **70**, and the second spacer **82**. In this embodiment, the first liner **68** and the second liner **100** are preferably made of different materials, in which the first liner **68** preferably includes silicon nitride (SiN) while the second liner **100** preferably includes silicon oxide or tetraethyl orthosilicate (TEOS). Nevertheless, according to other embodiments of the present invention, the two layers **68**, **100** could also be selected from the group consisting of SiN, SiO.sub.2, SiON, and SiCN while the two layers **68**, **100** are made of different materials. Moreover, the thickness of the first spacer **70** and/or second spacer **82** could be substantially the same as the thickness of the second liner **100**.

(18) Next, as shown in FIG. **5**, another IMD layer **72** is formed on the MTJ region **14** and logic region **16**, and a pattern transfer process is conducted by using a patterned mask (not shown) to remove part of the IMB layer **72** on the logic region **16** to form a contact hole (not shown) exposing the metal interconnection **26** underneath and metals are deposited into the contact hole afterwards. For instance, a barrier layer **34** selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and metal layer **36** selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP) could be deposited into the contact holes, and a planarizing process such as chemical mechanical polishing (CMP) process could be conducted to remove part of the metals including the aforementioned barrier layer and metal layer to form a contact plug **74** in the contact hole electrically connecting the metal interconnection **26**.

(19) Next, as shown in FIG. **6**, a stop layer **80** and another IMB layer **86** are formed on the MTJ **62** to cover the surface of the IMB layer **72**, and one or more photo-etching process is conducted to remove part of the IMD layer **86**, part of the stop layer **80**, part of the IMD layer **72**, part of the second liner **100**, and even part of the spacer adjacent to the MTJ **62** on the MTJ region **14** and part of the IMB layer **86** and part of the stop layer **80** on the logic region **16** to form contact holes (not shown). Next, conductive materials are deposited into each of the contact holes and a planarizing process such as CMP is conducted to form metal interconnections **88**, **90** directly connecting the MTJ **62** and contact plug **74** on the MTJ region **14** and logic region **16**, in which the metal interconnection **88** including protrusions **98** on the MTJ region **14** preferably directly contacts the MTJ **62** and/or left and right sidewalls of the top electrode **78** underneath while the metal interconnection **90** on the logic region **16** directly contacts the contact plug **74** on the lower level. Next, another stop layer **96** is formed on the IMB layer **86** to cover the metal interconnections **88**, **90**. In this embodiment, the width of the metal interconnection **88**, especially the via conductor of the metal interconnection **88** directly contacting the MTJ **62** is preferably greater than the width of the MTJ **62**, the bottom of each of the protrusions **98** or protruding portions include a planar surface, and the bottom surfaces of the protrusions **98** are preferably higher than the top surface of the capping layer **50**. Moreover, the bottom surface of each of the protrusions **98** contact the first spacer **70** and the second spacer **82** respectively, the sidewalls of each of the protrusions **98** are aligned with sidewalls of the first spacer **70** and the second spacer **82**, and the thickness or width of each of the first spacer **70** and the second spacer **82** could be less than or equal to the thickness or width of the second liner **100**.

(20) In this embodiment, the stop layer **80** and the stop layer **28** could be made of same material or different material. For example, both layers **80**, **28** could include nitrogen doped carbide (NDC), silicon nitride, silicon carbon nitride (SiCN), or combination thereof. Similar to the metal

interconnections formed previously, each of the metal interconnections **88**, **90** could be formed in the IMD layer **86** through a single damascene or dual damascene process. For instance, each of the metal interconnections **88**, **90** could further include a barrier layer **92** and a metal layer **94**, in which the barrier layer **92** could be selected from the group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), and tantalum nitride (TaN) and the metal layer **94** could be selected from the group consisting of tungsten (W), copper (Cu), aluminum (Al), titanium aluminide (TiAl), and cobalt tungsten phosphide (CoWP). Since single damascene process and dual damascene process are well known to those skilled in the art, the details of which are not explained herein for the sake of brevity. This completes the fabrication of a semiconductor device according to an embodiment of the present invention.
(21) Referring to FIG. **7**, FIG. **7** illustrates a structural view of a semiconductor device according to an embodiment of the present invention. As shown in FIG. **7**, in contrast to the bottom surfaces of the protrusions **68** directly contacting the top surfaces of the first spacer **70** and the second spacer **82** while sidewalls of each of the protrusions **98** are aligned with sidewalls of the first spacer **70** and second spacer **82** respectively as shown in FIG. **6**, it would also be desirable to width of the protrusions **98** such that the bottom surfaces of the protrusions **98** directly contact the top surface of the first spacer **70**, the top surface of the second spacer **82**, and the top surface of the second liner **100** while sidewalls of the protrusions **98** are aligned with sidewalls of the second liner **100**, which is also within the scope of the present invention.
(22) Referring to FIG. **8**, FIG. **8** illustrates a structural view of a semiconductor device according to an embodiment of the present invention. As shown in FIG. **8**, in contrast to the top electrode **78** being made of tantalum (Ta) as shown in FIG. **6**, the top electrode **78** of this embodiment is preferably made of titanium nitride (TiN) so that when the second electrode layer **52** is patterned by etching process to form the MTJ **62** in FIG. **2** the top electrode **78** of the MTJ **62** preferably includes a planar top surface and planar and vertical sidewalls. Next, fabrications illustrated in FIGS. **3-6** are conducted to form a first liner **62** on sidewalls of the MTJ **62**, remove part of the first liner **68** to form a first spacer **70** and second spacer **82** adjacent to the MTJ **62**, form a second liner **100** to cover the IMD layer **30**, the MTJ **62**, the first spacer **70**, and the second spacer **82**, form an IMD layer **72**, a stop layer **80**, and another IMD layer **86** on the MTJ **62**, and form metal interconnections **88**, **90** connecting the MTJ **62** and metal interconnection **72**. Preferably, a sidewall of the first spacer **70** directly connects a sidewall of the metal interconnection **32** and a sidewall of the second spacer **82** directly connects a sidewall of the metal interconnection **32**.
(23) Referring to FIG. **9**, FIG. **9** illustrates a structural view of a semiconductor device according to an embodiment of the present invention. As shown in FIG. **9**, it would also be desirable to first form the first liner **68** as shown in FIG. **3**, skip the etching process conducted in FIG. **4**, and then directly form a second liner **100** on the surface of the first liner **68**. Next, processes conducted in FIGS. **5-6** could be carried out to form an IMD layer **72**, a stop layer **80**, and another IMD layer **86** on the second liner **100**, and finally form metal interconnections **88**, **90** to connect the MTJ **62** and the metal interconnection **74** respectively.
(24) It should be noted since the first liner **68** has not been etched to form spacers in this embodiment, the first liner **68** itself after being deposited on the surface of the IMD layer **30** and MTJ **62** preferably include uneven thickness. For instance, the portion of the first liner **68** disposed on or directly contacting sidewalls of the MTJ **62** and the portion of the first liner **68** disposed on or directly contacting the top surface of the IMD layer **30** preferably include different thicknesses, in which the thickness of the first liner **68** directly contacting sidewalls of the MTJ **62** is preferably less than the thickness of the first liner **68** directly contacting the top surface of the IMD layer **30**. In this embodiment, the thickness of the first liner **68** disposed on the sidewalls of the MTJ **62** is preferably between 5-30 nm while the thickness of the first liner **68** disposed on the top surface of the IMD layer **30** is between 6-40 nm. In contrast to the first liner **68** having uneven thickness, the second liner **100** disposed on the surface of the first liner **68** preferably includes an even thickness

while the thickness of the second liner **100** is preferably equal to the thickness of the first liner **68** disposed on sidewalls of the MTJ **62** or between 5-30 nm.

(25) Referring to FIG. **10**, FIG. **10** illustrates a structural view of a semiconductor device according to an embodiment of the present invention. As shown in FIG. **10**, it would be desirable to form a top electrode **78** made of TiN as shown in FIG. **8**, and then follow the embodiment as disclosed in FIG. **9** to form a first liner **68** according to FIG. **3**, skip the etching process conducted in FIG. **4**, and then directly form a second liner **100** on the surface of the first liner **68**. Next, processes conducted in FIGS. **5-6** could be carried out to form an IMD layer **72**, a stop layer **80**, and another IMD layer **86** on the second liner **100**, and finally form metal interconnections **88**, **90** to connect the MTJ **62** and the metal interconnection **74** respectively.

(26) Similar to the embodiment disclosed in FIG. **9**, since the first liner **68** has not been etched to form spacers in this embodiment, the first liner **68** itself after being deposited on the surface of the IMD layer **30** and MTJ **62** preferably include uneven thickness. For instance, the portion of the first liner **68** disposed on or directly contacting sidewalls of the MTJ **62** and the portion of the first liner **68** disposed on or directly contacting the top surface of the IMD layer **30** preferably include different thicknesses, in which the thickness of the first liner **68** directly contacting sidewalls of the MTJ **62** is preferably less than the thickness of the first liner **68** directly contacting the top surface of the IMD layer **30**. In this embodiment, the thickness of the first liner **68** disposed on the sidewalls of the MTJ **62** is preferably between 5-30 nm while the thickness of the first liner **68** disposed on the top surface of the IMD layer **30** is between 6-40 nm. In contrast to the first liner **68** having uneven thickness, the second liner **100** disposed on the surface of the first liner **68** preferably includes an even thickness while the thickness of the second liner **100** is preferably equal to the thickness of the first liner **68** disposed on sidewalls of the MTJ **62** or between 5-30 nm.

(27) Referring to FIG. **11**, FIG. **11** illustrates a structural view of a semiconductor device according to an embodiment of the present invention. As shown in FIG. **11**, it would be desirable to combine the embodiments from FIG. **7** and FIG. **9** by first forming a first liner **68** according to FIG. **3**, skip the etching process conducted in FIG. **4**, and then directly form a second liner **100** on the surface of the first liner **68**. Next, processes conducted in FIGS. **5-6** could be carried out to form an IMD layer **72**, a stop layer **80**, and another IMD layer **86** on the second liner **100**, and finally form metal interconnections **88**, **90** to connect the MTJ **62** and the metal interconnection **74** respectively.

(28) Moreover, similar to FIG. **7**, in contrast to the bottom surface of the protrusions **98** contacting the top surfaces of the first spacer **70** and the second spacer **82** while sidewalls of the protrusions **98** are aligned with sidewalls of the first spacer **70** and second spacer **82**, it would be desirable to adjust the width of the metal interconnection **88** so that the bottom surfaces of the protrusions **98** contact the top surfaces of the first spacer **70**, second spacer **82**, and second liner **100** while sidewalls of the protrusions **98** are aligned with sidewalls of the second liner **100**, which is also within the scope of the present invention.

(29) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor device, comprising: a magnetic tunneling junction (MTJ) on a substrate; a first spacer on one side of the of the MTJ; a second spacer on another side of the MTJ; a first metal interconnection on the MTJ; a liner adjacent to the first spacer, the second spacer, and the first metal interconnection, wherein the liner directly contacts a sidewall of the first metal interconnection; a second metal interconnection under the MTJ, wherein a bottommost tip of a sidewall of the first spacer directly contacts a topmost tip of a sidewall of the second metal

interconnection; a first inter-metal dielectric (IMD) layer around the liner; and a second IMD layer around the second metal interconnection, wherein the bottommost tip of the sidewall of the first spacer that directly contacts the topmost tip of the sidewall of the second metal interconnection also directly contacts a topmost tip of a sidewall of the second IMD layer.

2. The semiconductor device of claim 1, wherein a top surface of the MTJ comprises a planar surface.

3. The semiconductor device of claim 1, wherein a bottom surface of the first metal interconnection comprises a planar surface.

4. The semiconductor device of claim 1, wherein a first sidewall of the first metal interconnection is aligned with a first sidewall of the MTJ.

5. The semiconductor device of claim 1, wherein a second sidewall of the first metal interconnection is aligned with a second sidewall of the MTJ.
